

Exploring the interaction between compact binaries and their host galaxy

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Aim of the Project

- 1 How different morphologies of galaxies effect binary orbits?
- 2 Where Binary systems are likely to exist? Does a relation exist between location and some of the binary properties?
- 3 What are the consequences of Stellar evolution?

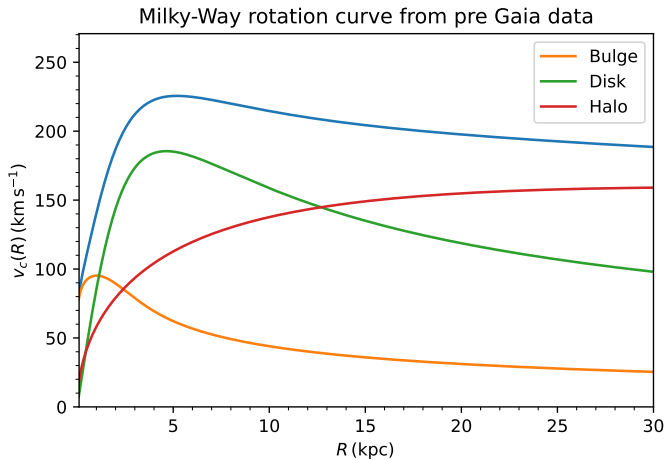
Introduction and Overview

- programming language used: Python
- **galpy** package used for galactic potentials and orbit integration
- simplistic version/model of Soheb's work
- currently analysing double neutron star systems
- extending the analysis to spiders

Milky Way Potential

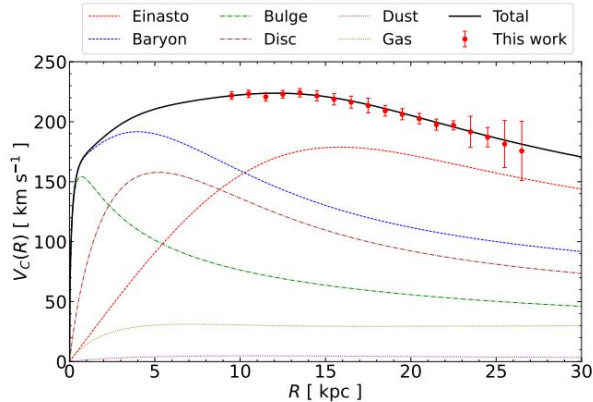
- Focused on analytic potential profiles
- Azimuthal symmetry for simplicity
- Implemented a description of a Milky Way potential from experimental data before Gaia
- Implement most recent rotation curve from Gaia DR3

Milky Way Potential



Most recent Milky Way Potential

Rotation curve from Gaia DR3 [1]



Most recent Milky Way Potential

- Keplerian decline in the rotational velocity from 15kpc
- Last data point at 27 kpc from galactic center
- Places an upper bound on the mass of our galaxy, which is less than previously thought
- Compare binary evolution in different models of Milky Way potentials

Creating an Initial Binary population

- Obtained a variety of binary models from BPASS [2]
- Neutron Star - Neutron Star systems (systems with 2 natal kicks)
- Seeding them into the galaxy using a PDF
- Baryonic matter mass distribution used as PDF for initial distribution

Evolving Orbits

- Stars placed at different radii generated from PDF
- Assumed initial circular orbits until the 1st natal kick
- After 2nd natal kick evolved until Hubble time
- Kicks applied in an overall isotropic fashion
- Merger times tracked, time and place of merges recorded

Classifying Orbits

- Orbits can be categorized according to different variables
- Bound and Unbound orbits
- In- or out of plane orbits
- Regular, irregular orbits
- Phase space and Integrals of motion used
- Machine Learning approach of orbit classification

Motivation behind project and results

- Could predict properties of a natal kick(s) from a merger location
- Investigate possible trends for location of binaries after natal kicks
- Been optimizing my code so currently running the integration
- Results will be analyzed hopefully before end of week

References

- 1 Jiao, Y., “Detection of the Keplerian decline in the Milky Way rotation curve”, *Astronomy and Astrophysics*, vol. 678, 2023. doi:10.1051/0004-6361/202347513.
- 2 Eldridge, J. J., Stanway, E. R. 2016, *MNRAS*, 462, 3302

Thank you!